

SUPPLEMENTARY MATERIAL FOR: THE COST OF UNKNOWN FRICTION IN MINIMUM-TIME VEHICLE TRANSIT

HIMAGHNA ROY CHOUDHURY

This supplement accompanies the paper *The Cost of Unknown Friction in Minimum-Time Vehicle Transit*. Section S1 gives the values behind every numerical statement of the paper, and Section S2 gives the proofs of the results of Section 15 of the paper. Theorems, sections, and equations without the prefix S refer to the paper.

S1. NUMERICAL EVIDENCE

In this section we give the values behind every numerical statement of the paper. Each table is generated by the code that accompanies the paper from the same computations that the text describes, so that every reported check can be inspected and reproduced.

S1.1. Measured data. Table S1 summarises the inversion of the EPA road-load coefficients. Tables S2–S5 give the D3 results of Section 3, and Table S6 the 0–60 mph comparison. Table S8 lists all 114 determinations of friction from the FMVSS 135 reports.

TABLE S1. Road-load coefficients of the 1680 vehicle configurations of model year 2013 in the EPA test car list, inverted with $C = \frac{1}{2}\rho C_D A$ and $A = C_{rr}mg$. The drag area lies in $[0.4, 2.2] \text{ m}^2$ for 98.2% of them and the rolling resistance coefficient in $[0.006, 0.015]$ for 91.8%. The median drag area is 0.699 m^2 for the 1061 configurations listed as cars, 1.064 m^2 for the 378 listed as trucks, and 0.766 m^2 for the 241 listed as both. A line through the origin fitted to the constant term A against the weight mg has slope 0.00874, correlation $r = 0.62$, and $R^2 = 0.37$.

Quantity	5%	25%	Median	75%	95%
Drag area $C_D A$ (m^2)	0.505	0.659	0.752	0.909	1.260
Rolling resistance C_{rr}	0.0056	0.0074	0.0087	0.0102	0.0124

TABLE S2. Mass recovered from the D3 trajectories. In each run, $F - R$ is regressed on the acceleration and the fitted mass is compared with the published test weight. The columns give the mean and the largest absolute error over the runs of each vehicle.

Vehicle	Type	Runs	Samples	Mean	Largest
2012 Nissan Leaf	BEV	15	120,277	−0.10%	0.34%
2010 Toyota Prius	HEV	5	52,921	+0.29%	0.72%
2012 Ford Focus	ICE	5	52,688	−0.03%	0.28%
2013 Ford Focus Electric	BEV	20	214,969	+0.01%	0.41%

TABLE S3. Information about the road-load coefficients by drive cycle. The table gives the smallest eigenvalue of the scaled normal matrix for (m, A, B, C) , averaged over the runs of each cycle, and its ratio to that of the urban cycle (UDDS).

Cycle	Runs	Smallest eigenvalue	Relative to UDDS
UDDS	21	9.49×10^{-3}	1.000
SC03 Prep, Ph 1	1	6.71×10^{-3}	0.708
US06	8	3.37×10^{-3}	0.355
Steady state	7	2.77×10^{-3}	0.291
Highway	6	9.62×10^{-4}	0.101

TABLE S4. Paired steady-state runs of the 2013 Ford Focus Electric on a level roll and at a simulated grade of 6%. The offset is the mean of the measured force less the predicted road load, in newtons. The difference of the offsets is 1051.79 N, against $mg \sin(\arctan 0.06) = 1051.80$ N.

Test	Grade	Offset (N)
61408008	0%	+229.0
61408009	6%	−822.8

TABLE S5. Pack resistance of the 2012 Nissan Leaf measured in D3 runs at two cell temperatures, from the regression of terminal voltage on current. An Arrhenius law through the two points gives a ratio of 1.66 between the resistances at 0°C and 25°C.

Cell temperature (°C)	Resistance (mΩ)
−6.4	183.5
21.2	102.8

TABLE S6. Predicted and published 0–60 mph times (s). Published times in the United States are measured after a one-foot rollout, so the prediction is compared under the same convention. The error is that of the rollout prediction, and the last column is the share of the predicted run during which traction binds. The published times are those of EV Specifications for the Leaf [2], the MotorWeek road test for the Civic [3], and the Corvette Action Center for the ZR1 [1].

Vehicle	Standing start	Rollout	Published	Error	Traction-limited
Leaf	10.57	10.16	9.90	+2.7%	0%
Civic	9.96	9.55	9.50	+0.5%	48%
ZR1	3.58	3.30	3.30	−0.2%	100%

Report	Vehicle	Load	a	t	μ_{eff}	PFC	Error
--------	---------	------	-----	-----	--------------------	-----	-------

Table S8: The 114 determinations of effective friction from the FMVSS 135 compliance reports. For each vehicle and loading, the stopping distances at two speeds determine the deceleration a (m/s^2) and the brake application delay t (s), and $\mu_{\text{eff}} = a/g$. PFC is the peak friction coefficient of the test surface measured with the reference tyre, and the last column is the hold-out error (%) of the predicted high-speed stop. Loading LLVW is lightly loaded and GVWR fully loaded.

Report	Vehicle	Load	a	t	μ_{eff}	PFC	Error
TRTR-643634-2014-001	2014 Acura RLX	GVWR	9.39	+0.227	0.957	0.94	+0.9
TRTR-643634-2014-001	2014 Acura RLX	LLVW	9.12	+0.079	0.930	0.94	-2.8
TRTR-641763-2011-001	2011 BMW 128I	GVWR	9.56	+0.271	0.974	0.97	+1.9
TRTR-641763-2011-001	2011 BMW 128I	LLVW	10.32	+0.264	1.053	0.97	+1.9
TRTR-644580-2016-001	2016 BMW X1	GVWR	9.41	+0.274	0.960	0.97	+2.0
TRTR-644580-2016-001	2016 BMW X1	LLVW	9.61	+0.170	0.979	0.97	-0.6
TRTR-644567-2016-001	2016 Buick Cascada	GVWR	8.71	+0.026	0.889	0.97	-3.9
TRTR-644567-2016-001	2016 Buick Cascada	LLVW	9.07	+0.046	0.925	0.97	-3.6
TRTR-647132-2020-001	2020 Buick Encore	LLVW	8.79	+0.049	0.896	0.95	-2.5
TRTR-644086-2015-001	2015 Cadillac ATS	GVWR	9.66	+0.096	0.985	0.92	-2.5
TRTR-645180-2017-001	2017 Cadillac XT5	GVWR	10.05	+0.260	1.025	0.96	+1.8
TRTR-645180-2017-001	2017 Cadillac XT5	LLVW	10.50	+0.160	1.071	0.96	-0.9
TRTR-641656-2011-001	2011 Chevrolet Cruze	GVWR	8.85	+0.216	0.902	0.94	+0.5
TRTR-641656-2011-001	2011 Chevrolet Cruze	LLVW	9.80	+0.285	0.999	0.94	+1.9
135-2400923-TEST	2024 Chevrolet Equinox	GVWR	9.79	+0.304	0.998	0.95	+2.3
135-2400923-TEST	2024 Chevrolet Equinox	LLVW	9.79	+0.152	0.999	0.95	-0.9
TRTR-643542-2014-001	2014 Chevrolet Silverado 1500 Crew CAB	GVWR	7.81	+0.123	0.797	0.96	-1.2
TRTR-643542-2014-001	2014 Chevrolet Silverado 1500 Crew CAB	LLVW	8.48	+0.144	0.864	0.96	-0.9
TRTR-642228-2012-001	2012 Chevrolet Sonic	GVWR	8.27	+0.040	0.843	0.96	-3.0
TRTR-642228-2012-001	2012 Chevrolet Sonic	LLVW	9.20	+0.164	0.938	0.96	-0.6
TRTR-643151-2013-001	2013 Chevrolet Spark	GVWR	9.57	+0.187	0.976	0.97	-0.1
TRTR-643151-2013-001	2013 Chevrolet Spark	LLVW	11.64	+0.405	1.187	0.97	+3.6
TRTR-644227-2015-001	2015 Chrysler 200	GVWR	11.33	+0.510	1.156	0.99	+7.8
TRTR-647131-2020-001	2020 Chrysler Pacifica Hybrid	GVWR	9.68	+0.280	0.987	0.95	+1.5
TRTR-647131-2020-001	2020 Chrysler Pacifica Hybrid	LLVW	10.21	+0.261	1.041	0.95	+1.3

Report	Vehicle	Load	a	t	μ_{eff}	PFC	Error
TRTR-642114-2012-001	2012 Dodge Du-rango	GVWR	9.42	+0.458	0.960	0.96	+4.7
TRTR-642114-2012-001	2012 Dodge Du-rango	LLVW	10.72	+0.457	1.093	0.96	+5.2
TRTR-643617-2014-001	2014 Fiat 500 L	GVWR	8.63	+0.040	0.880	0.94	-3.5
TRTR-643617-2014-001	2014 Fiat 500 L	LLVW	9.67	+0.112	0.986	0.94	-2.0
135-2360678-TEST	2023 Ford Bronco	GVWR	7.48	+0.045	0.763	1.09	-1.8
135-2360678-TEST	2023 Ford Bronco	LLVW	7.36	-0.041	0.750	1.09	-2.9
TRTR-647510-2021-001	2022 Ford Bronco Sport	GVWR	9.18	+0.019	0.936	0.94	-4.0
TRTR-647510-2021-001	2022 Ford Bronco Sport	LLVW	10.22	+0.176	1.042	0.94	-0.5
TRTR-643505-2013-001	2013 Ford C-MAX Energi	GVWR	9.26	+0.196	0.944	0.97	+0.2
TRTR-643505-2013-001	2013 Ford C-MAX Energi	LLVW	9.66	+0.208	0.985	0.97	+0.4
TRTR-639664-2008	2008 Ford Escape	GVWR	10.06	+0.318	1.026	1.04	+2.0
TRTR-639664-2008	2008 Ford Escape	LLVW	9.25	+0.212	0.943	1.04	+0.4
TRTR-642375-2012-001	2012 Ford Mustang	GVWR	7.65	-0.077	0.780	0.97	-5.8
TRTR-642375-2012-001	2012 Ford Mustang	LLVW	10.27	+0.382	1.047	0.97	+4.6
TRTR-646191-2019-001	2019 Ford Ranger	GVWR	10.08	+0.483	1.027	1.04	+5.0
TRTR-646191-2019-001	2019 Ford Ranger	LLVW	11.23	+0.576	1.145	1.04	+6.9
TRTR-641637-2011-001	2011 Ford Transit Connect	LLVW	10.40	+0.388	1.060	0.98	+1.2
TRTR-641694-2011-001	2011 Honda Civic	GVWR	9.97	+0.548	1.017	0.97	+7.0
TRTR-641694-2011-001	2011 Honda Civic	LLVW	8.46	+0.087	0.863	0.97	-2.2
TRTR-644067-2015-001	2015 Honda Odyssey	GVWR	8.80	+0.156	0.897	0.96	-0.8
TRTR-642537-2012-001	2012 Honda Pilot	GVWR	9.59	+0.466	0.978	0.97	+4.9
TRTR-642537-2012-001	2012 Honda Pilot	LLVW	8.91	+0.251	0.908	0.97	+1.1
TRTR-639189-2006	2008 Hummer H3	GVWR	7.24	-0.054	0.738	1.03	-2.7
TRTR-639189-2006	2008 Hummer H3	LLVW	7.65	-0.105	0.780	1.03	-3.5
135-2602381-TEST	2026 Hyundai Santa FE	GVWR	7.60	-0.179	0.775	1.16	-7.5
135-2602381-TEST	2026 Hyundai Santa FE	LLVW	10.29	+0.228	1.049	1.16	+0.9
TRTR-644174-2015-001	2015 Hyundai Sonata SE	GVWR	9.00	+0.073	0.918	0.96	-2.9
TRTR-642982-2013-001	2013 Hyundai Tucson	GVWR	8.29	+0.082	0.846	1.00	-1.9
TRTR-642982-2013-001	2013 Hyundai Tucson	LLVW	8.93	+0.091	0.911	1.00	-1.9
TRTR-642374-2012-001	2012 Hyundai Veloster	GVWR	11.85	+0.768	1.208	0.95	+10.2
TRTR-642374-2012-001	2012 Hyundai Veloster	LLVW	10.47	+0.427	1.068	0.95	+4.4
TRTR-646752-2020-001	2020 Jeep Gladiator	GVWR	8.14	+0.197	0.830	0.95	+0.2
TRTR-646752-2020-001	2020 Jeep Gladiator	LLVW	10.02	+0.334	1.021	0.95	+2.7

Report	Vehicle	Load	a	t	μ_{eff}	PFC	Error
TRTR-643638-2014-001	2014 Jeep Grand Cherokee	GVWR	8.04	+0.177	0.820	0.94	-0.4
TRTR-644557-2016-001	2016 Jeep Renegade	GVWR	8.95	+0.197	0.912	0.96	+0.2
TRTR-644557-2016-001	2016 Jeep Renegade	LLVW	9.59	+0.226	0.978	0.96	+0.8
135-2360682-TEST	2023 Jeep Wagoneer	GVWR	9.06	+0.260	0.924	1.09	+1.4
135-2360682-TEST	2023 Jeep Wagoneer	LLVW	9.53	+0.264	0.972	1.09	+1.6
TRTR-643637-2014-001	2014 KIA Cadenza	GVWR	9.09	+0.132	0.927	0.94	-1.5
TRTR-647598-2021-001	2021 KIA K5	GVWR	9.40	-0.001	0.958	1.01	-5.0
TRTR-647598-2021-001	2021 KIA K5	LLVW	10.17	+0.050	1.037	1.01	-3.9
TRTR-641847-2011-001	2011 KIA Soul	GVWR	7.38	-0.004	0.752	0.97	-2.8
TRTR-641847-2011-001	2011 KIA Soul	LLVW	8.62	+0.178	0.879	0.97	-0.3
TRTR-647204-2020-001	2021 KIA Telluride	GVWR	9.00	+0.006	0.918	0.95	-4.6
TRTR-647204-2020-001	2021 KIA Telluride	LLVW	10.30	+0.123	1.050	0.95	-1.9
TRTR-641781-2011-001	2011 Mazda 2	GVWR	12.60	+0.762	1.285	0.97	+6.9
TRTR-641781-2011-001	2011 Mazda 2	LLVW	10.66	+0.461	1.087	0.97	+3.4
TRTR-639212-2006-001	2008 Mazda CX-7	GVWR	8.73	+0.212	0.890	1.01	+0.5
TRTR-639212-2006-001	2008 Mazda CX-7	LLVW	8.94	-0.060	0.911	1.01	-5.8
TRTR-643633-2014-001	2014 MERCEDES-BENZ CLA 250	GVWR	10.05	+0.332	1.025	0.94	+3.4
TRTR-643633-2014-001	2014 MERCEDES-BENZ CLA 250	LLVW	9.63	+0.102	0.982	0.94	-2.3
135-2400158-TEST	2023 MERCEDES-BENZ EQS-CLASS Sedan	GVWR	11.88	+0.458	1.212	0.95	+7.0
135-2400158-TEST	2023 MERCEDES-BENZ EQS-CLASS Sedan	LLVW	10.05	+0.071	1.025	0.95	-3.3
TRTR-647264-2020-001	2020 Mitsubishi Outlander	LLVW	10.63	+0.237	1.084	0.96	+0.9
TRTR-639241-2006	2008 Nissan Altima	GVWR	9.37	+0.332	0.955	1.04	+2.7
TRTR-639241-2006	2008 Nissan Altima	LLVW	12.78	+0.559	1.303	1.04	+8.0
TRTR-647267-2020-001	2021 Nissan Frontier	LLVW	9.87	+0.612	1.006	0.95	+7.1
135-2400952-TEST	2025 Nissan Kicks	GVWR	9.04	+0.086	0.922	1.10	-2.1
TRTR-644558-2016-001	2016 Nissan Maxima	GVWR	9.54	+0.157	0.973	0.96	-0.9
TRTR-644558-2016-001	2016 Nissan Maxima	LLVW	9.42	+0.110	0.961	0.96	-2.1
135-2360684-TEST	2024 Nissan Pathfinder	GVWR	9.62	+0.250	0.981	1.09	+1.4
135-2360684-TEST	2024 Nissan Pathfinder	LLVW	10.52	+0.351	1.072	1.09	+3.8

Report	Vehicle	Load	a	t	μ_{eff}	PFC	Error
TRTR-639190-2006	2008 Saturn Out-look	GVWR	8.90	+0.305	0.908	1.01	+2.0
TRTR-639190-2006	2008 Saturn Out-look	LLVW	7.63	-0.147	0.778	1.01	-5.9
TRTR-642113-2012-001	2012 Scion TC	GVWR	7.99	+0.011	0.815	0.95	-3.8
TRTR-642113-2012-001	2012 Scion TC	LLVW	8.91	+0.108	0.909	0.95	-1.9
TRTR-643210-2013-001	2013 Subaru Tribeca	GVWR	7.92	-0.083	0.808	0.97	-6.1
TRTR-643210-2013-001	2013 Subaru Tribeca	LLVW	9.55	+0.173	0.974	0.97	-0.5
TRTR-647265-2020-001	2020 Subaru WRX	LLVW	9.96	+0.078	1.016	0.99	-3.1
135-2360686-TEST	2023 Tesla Model Y	GVWR	11.92	+0.329	1.216	1.13	+3.9
135-2360686-TEST	2023 Tesla Model Y	LLVW	11.20	+0.224	1.142	1.13	+1.0
TRTR-641858-2011-001	2011 Toyota Camry	GVWR	9.35	+0.289	0.953	0.98	+2.3
TRTR-641858-2011-001	2011 Toyota Camry	LLVW	9.84	+0.281	1.004	0.98	+2.2
TRTR-644080-2015-001	2015 Toyota Camry	GVWR	9.24	+0.218	0.942	0.92	+0.6
135-2501546-TEST	2025 Toyota Prius	GVWR	9.35	+0.140	0.954	1.14	-1.1
135-2501546-TEST	2025 Toyota Prius	LLVW	10.83	+0.384	1.104	1.14	+4.1
TRTR-646190-2019-001	2019 Toyota RAV4	LLVW	9.34	+0.061	0.953	1.04	-3.2
TRTR-641860-2011-001	2011 Toyota Tundra	GVWR	8.24	+0.302	0.840	0.97	+1.8
TRTR-641860-2011-001	2011 Toyota Tundra	LLVW	9.01	+0.323	0.919	0.97	+2.3
TRTR-647600-2021-001	2022 Volkswagen Atlas	GVWR	10.36	+0.136	1.057	0.94	-1.4
TRTR-647600-2021-001	2022 Volkswagen Atlas	LLVW	10.27	+0.122	1.048	0.94	-1.7
TRTR-643668-2014-001	2014 Volkswagen CC	GVWR	9.05	+0.178	0.923	0.95	-0.4
TRTR-644178-2015-001	2015 Volkswagen Golf	GVWR	9.65	+0.234	0.984	0.94	+1.1
TRTR-641695-2011-001	2011 Volkswagen Jetta	GVWR	11.27	+0.567	1.149	0.97	+7.4
TRTR-641695-2011-001	2011 Volkswagen Jetta	LLVW	13.24	+0.629	1.351	0.97	+9.4
135-2602379-TEST	2026 Volkswagen Taos	GVWR	8.23	+0.020	0.840	1.16	-3.9
135-2602379-TEST	2026 Volkswagen Taos	LLVW	9.01	+0.121	0.918	1.16	-1.7
TRTR-644652-2016-001	2016 Volvo XC90	GVWR	9.04	+0.008	0.922	0.97	-4.6
TRTR-644652-2016-001	2016 Volvo XC90	LLVW	10.02	+0.122	1.022	0.97	-1.9

TABLE S7. Statistics of the FMVSS 135 determinations listed in Table S8. The hold-out error predicts the high-speed stop of each vehicle from its low-speed stop and the median delay of all other vehicles, and compares the prediction with the measured distance.

Quantity	Value
Determinations of friction	114
Brake application delay t , median (interquartile range)	0.192 s (0.086 to 0.304)
Effective friction μ_{eff} , median (interquartile range)	0.967 (0.909 to 1.027)
Hold-out error of the high-speed stop, mean absolute	2.82%
Hold-out error, mean (standard deviation)	+0.27% (3.56%)
Hold-out errors within 10%	99% of 114
Vehicles tested at both loadings	49
Laden friction lower than lightly loaded	36 of 49
Relative change of friction when laden, median	−7.17%
Two-sided sign test	$p = 1.4 \times 10^{-3}$
Correlation of μ_{eff} with PFC	$r = -0.031$

S1.2. **Well-posedness.** Tables S9 and S10 give the values behind the check of Theorem 4.1 in Section 4.

TABLE S9. Check of Theorem 4.1. For each vehicle, tyre law, and speed, the effective mass m_{eff} and the load-transfer gain g (both in kg), the margin $m_{\text{eff}} - g$, the largest slope of the force margin Φ on a grid of 4001 accelerations (in N per m/s^2 , negative when Φ is strictly decreasing), and the envelope a_{max} (m/s^2). A negative envelope means that the vehicle cannot sustain that speed.

Vehicle	Tyre law	mph	m_{eff}	g	$m_{\text{eff}} - g$	$\max \Phi'$	a_{max}
Leaf	ellipse	5	1626	-269.0	1895.0	-1626	3.663
Leaf	ellipse	30	1626	-269.0	1895.0	-1626	3.154
Leaf	ellipse	60	1626	-269.0	1895.0	-1626	1.335
Leaf	ellipse	100	1626	-269.0	1895.0	-1626	-0.712
Leaf	PAC2002	5	1626	-157.2	1783.3	-1626	3.663
Leaf	PAC2002	30	1626	-157.2	1783.3	-1626	3.154
Leaf	PAC2002	60	1626	-157.2	1783.3	-1626	1.335
Leaf	PAC2002	100	1626	-157.2	1783.3	-1626	-0.712
Civic	ellipse	5	1625	-231.5	1856.5	-1625	3.567
Civic	ellipse	30	1625	-231.5	1856.5	-1625	3.519
Civic	ellipse	60	1397	-231.5	1628.8	-1397	1.629
Civic	ellipse	100	1379	-231.5	1610.1	-1379	0.761
Civic	PAC2002	5	1625	-135.5	1760.5	-1625	3.531
Civic	PAC2002	30	1625	-135.5	1760.5	-1625	3.484
Civic	PAC2002	60	1397	-135.5	1532.8	-1397	1.629
Civic	PAC2002	100	1379	-135.5	1514.1	-1379	0.761
ZR1	ellipse	5	1784	361.3	1423.2	-1423	7.591
ZR1	ellipse	30	1784	361.3	1423.2	-1423	7.503
ZR1	ellipse	60	1784	361.3	1423.2	-1423	7.328
ZR1	ellipse	100	1681	361.3	1320.2	-1320	4.875
ZR1	PAC2002	5	1784	410.1	1374.3	-1374	7.211
ZR1	PAC2002	30	1784	410.1	1374.3	-1374	7.123
ZR1	PAC2002	60	1784	410.1	1374.3	-1374	6.948
ZR1	PAC2002	100	1681	410.1	1271.4	-1271	4.875

TABLE S10. Check of the bound of Theorem 4.1(iii) at 10 mph. The measured value is $|\partial a_{\text{max}}/\partial \mu|$ from a relative change of 10^{-4} in friction, and the bound is $|\partial_{\mu} \Phi|/(m_{\text{eff}} - g)$, both in m/s^2 .

Vehicle	Tyre law	Binding	Measured	Bound
Leaf	ellipse	power	0.0000	0.0000
Leaf	PAC2002	power	0.0000	0.0000
Civic	ellipse	traction	3.7368	3.7369
Civic	PAC2002	traction	3.7818	3.9060
ZR1	ellipse	traction	7.1418	7.1416
ZR1	PAC2002	traction	6.4143	7.0277

S1.3. **Monotonicity and parameter signs.** Tables S11 and S12 give the values behind the check of Theorem 6.2, and Tables S13 and S14 those behind the check of Theorem 6.4.

TABLE S11. Check of Theorem 6.2(i) and (ii). Each parameter was swept over the stated range of multipliers in 11 steps, and grade in 9 steps, for the three vehicles, both tyre laws, and speeds of 10, 40, and 80 mph. The second column gives the predicted direction of $a_{\max}$ as the parameter increases, and a pair is two consecutive points of a sweep. The last column is the smallest change of $a_{\max}$ (m/s²) in the predicted direction over all pairs, which is non-negative when no pair violates the prediction. It is zero, to rounding error below 10^{-12} , for friction and for power, because the envelope does not respond to friction where power binds or to power where traction binds (Proposition 5.3). The largest value of $\mu_r \varphi'_r h/L$ encountered in the grade sweeps was 0.227.

Parameter	Predicted	Range	Pairs	Violations	Smallest step
Friction μ	up	0.5–1.5	180	0	0
Rolling resistance C_{rr}	down	0.5–2.0	180	0	7.86×10^{-3}
Drag area $C_D A$	down	0.5–2.0	180	0	6.24×10^{-4}
Power P	up	0.5–1.5	180	0	0
Uphill grade	down	0°–8°	144	0	1.43×10^{-1}

TABLE S12. Check of Theorem 6.2(iii) and (iv). For each case the table gives the binding constraint, the terms $Ja_{\max}$, F_{aero} , and Λ (N) of the mass formula, the predicted sign and the finite-difference value of $\partial a_{\max}/\partial \ln m$, and the predicted sign and finite-difference value of $\partial a_{\max}/\partial \ln \rho$, both in m/s² from relative changes of 10^{-4} . Where power binds, the mass prediction is negative by part (iii).

Vehicle	Law	mph	Binding	$Ja_{\max}$	F_{aero}	Λ	$\partial_{\ln m} a_{\max}$	$\partial_{\ln \rho} a_{\max}$		
Leaf	ellipse	10	power	271	10	n/a	−	−3.584	−	−0.0060
Leaf	ellipse	40	power	170	157	n/a	−	−2.275	−	−0.0966
Leaf	ellipse	80	power	56	628	n/a	−	−0.817	−	−0.3862
Leaf	PAC2002	10	power	271	10	n/a	−	−3.584	−	−0.0060
Leaf	PAC2002	40	power	170	157	n/a	−	−2.275	−	−0.0966
Leaf	PAC2002	80	power	56	628	n/a	−	−0.817	−	−0.3862
Civic	ellipse	10	traction	1080	8	0	+	0.586	−	−0.0042
Civic	ellipse	40	power	342	124	n/a	−	−2.495	+	2.7308
Civic	ellipse	80	power	107	494	n/a	−	−1.450	+	1.5265
Civic	PAC2002	10	traction	1070	8	866	+	0.117	−	−0.0042
Civic	PAC2002	40	power	342	124	n/a	−	−2.495	+	2.7308
Civic	PAC2002	80	power	107	494	n/a	−	−1.450	+	1.5265
ZR1	ellipse	10	traction	1437	7	0	+	1.015	−	−0.0052
ZR1	ellipse	40	traction	1414	119	0	+	1.077	−	−0.0835
ZR1	ellipse	80	power	765	475	n/a	−	−6.268	+	6.7155
ZR1	PAC2002	10	traction	1365	7	2468	−	−0.729	−	−0.0049
ZR1	PAC2002	40	traction	1343	119	2468	−	−0.670	−	−0.0791
ZR1	PAC2002	80	power	765	475	n/a	−	−6.268	+	6.7155

TABLE S13. Nestedness of the PAC2002 combined-slip force set under the friction scale $\lambda_\mu \in \{0.3, 0.5, 0.7, 0.9, 1.0, 1.2\}$, for three normal loads. The largest outward violation is the largest relative amount by which the boundary at a smaller scale lies outside the boundary at the next larger scale, over 72 directions. The last column is the range over directions of the ratio of the boundary radius at scale 1.2 to that at scale 0.3.

Load (F_{z0})	Largest outward violation	Radius ratio
0.5	0.00%	3.29–6.36
1.0	0.00%	3.26–6.41
1.5	0.00%	3.19–6.23

TABLE S14. Four-wheel equilibrium envelope $a_{\max}$ (m/s²) of the Corvette ZR1 at 20 m/s with PAC2002 combined slip, against a multiplier on friction.

Radius	$\times 0.6$	$\times 0.7$	$\times 0.8$	$\times 0.9$	$\times 1.0$	$\times 1.1$
straight	3.834	4.598	5.390	6.210	7.060	7.941
300 m	3.808	4.569	5.357	6.173	7.019	7.895
150 m	3.724	4.473	5.249	6.052	6.885	7.746
100 m	3.547	4.280	5.036	5.818	6.626	7.463
70 m	2.933	3.718	4.472	5.226	6.005	6.802

S1.4. **Identifiability.** Table S15 gives the bounds identified from ordinary driving, Tables S16 and S17 the Fisher ranks and standard errors of Theorem 7.5, and Tables S18 and S19 the values behind Theorem 7.6.

TABLE S15. Lower bounds on μ and on $P_{\max}$ identified from ordinary driving in the D3 data, over all drive cycles including US06, using the driving samples with $v > 1$ m/s and $F > 0$. The bound $F/(mg)$ needs no geometry. The driven-axle bound uses the nominal front weight fraction and $h/L = 0.2$, without and with 1 s averaging of the force. The last column is the largest delivered power as a fraction of the rated power, followed by the accelerator pedal position at that sample, where it is recorded. No sample slips or binds, so each value is a lower bound only. The total is 306,708 samples.

Vehicle	Samples	$F/(mg)$	Axle	Axle, 1 s	Power	Pedal
2012 Nissan Leaf	88,186	0.312	0.600	0.597	0.95	54%
2010 Toyota Prius	37,819	0.398	0.759	0.722	0.99	n/a
2012 Ford Focus	39,967	0.400	0.738	0.686	0.68	n/a
2013 Ford Focus Electric	140,736	0.317	0.636	0.629	0.81	78%

TABLE S16. Check of Theorem 7.5. Every design contains 60 balance samples at speeds drawn uniformly from 5 to 30 m/s and accelerations from -1.5 to 2.0 m/s². The limit events are five samples at 3.0 m/s² and 8 m/s, plus, for the two-acceleration design, five at 1.5 m/s² and 20 m/s. The braking events are five at -7 and five at -5 m/s². Each limit event contributes a balance row and a limit row. The singular values are those of the Jacobian with each parameter scaled to its nominal value.

Design	Rows	Rank	Singular values
Ordinary driving	60	4	1.5×10^4 , 2.3×10^3 , 8.1×10^2 , 2.3×10^1 , 5.3×10^{-7} , 0
Limit events at one acceleration	70	5	2.5×10^4 , 1.0×10^4 , 2.4×10^3 , 8.5×10^2 , 2.3×10^1 , 6.3×10^{-7}
Limit events at two accelerations	80	6	3.3×10^4 , 1.2×10^4 , 2.4×10^3 , 8.5×10^2 , 2.4×10^1 , 1.9×10^0
Braking-limit events	80	5	5.7×10^4 , 1.9×10^4 , 3.2×10^3 , 8.2×10^2 , 2.3×10^1 , 7.7×10^{-7}

TABLE S17. Cramér–Rao standard errors for the design with limit events at two accelerations, with force noise of 50 N on each of its 80 rows. The nominal values are those of a Civic-like car, with grade in radians and wind in m/s.

Parameter	Nominal	Relative error	Absolute error
μ	0.85	8.9%	0.0754
m	1400	0.4%	4.9
C_{rr}	0.01	2338.7%	0.234
$C_D A$	0.7	30.3%	0.212
γ	0.02	1169.5%	0.234
w	3	203.7%	6.11

TABLE S18. The exact friction sensitivity $x^2(3 - 2x)$ of Theorem 7.6(i), in units of μN , against its small-utilisation form $u^2/3$.

u	x	$x^2(3 - 2x)$	$u^2/3$	Ratio
0.05	0.0170	0.00085	0.00083	1.023
0.10	0.0345	0.00349	0.00333	1.047
0.20	0.0717	0.01468	0.01333	1.101
0.30	0.1121	0.03488	0.03000	1.163
0.50	0.2063	0.11012	0.08333	1.321
0.80	0.4152	0.37401	0.21333	1.753
0.95	0.6316	0.69284	0.30083	2.303

TABLE S19. Ratios of Cramér–Rao standard deviations of friction. The first column divides the deviation at $u_{\max} = 0.3$ by that at $u_{\max} = 0.8$, both with $n = 10$. The second divides the deviation at $u_{\max} = 0.5$ with $n = 160$ by that at $u_{\max} = 0.8$ with $n = 10$. The brush-model rows use the exact sensitivities with the stiffness estimated jointly.

Source	First ratio	Second ratio
Published by Jonasson et al.	16.53	1.259
Brush model, uniform in slip	17.16	1.325
Brush model, uniform in utilisation	16.57	1.313

S1.5. **Uncertainty propagation.** Tables S20–S22 give the values behind the check of Theorem 8.1.

TABLE S20. Check of Theorem 8.1(ii) at 10 mph. The friction distribution is the 114 values of Table S8 divided by their median, with relative standard deviation 0.120 and 5th and 95th percentiles 0.806 and 1.236. The table gives the exact quantiles $a_{\max}(q_{\mu}(p))$ (m/s²) at $p = 0.05, 0.25, 0.5, 0.75$, and 0.95 , and the largest difference from the Monte Carlo quantiles of the 114 envelope values.

Vehicle	Law	Exact quantiles	Largest difference
Leaf	ellipse	3.3899, 3.6670, 3.6670, 3.6670, 3.6670	4.4×10^{-16}
Leaf	PAC2002	3.3781, 3.6670, 3.6670, 3.6670, 3.6670	4.4×10^{-16}
Civic	ellipse	2.9279, 3.3723, 3.5458, 3.7627, 4.1616	0
Civic	PAC2002	2.8907, 3.3385, 3.5136, 3.7332, 4.1616	0
ZR1	ellipse	5.7904, 7.0162, 7.5298, 8.2029, 9.7709	0
ZR1	PAC2002	5.5650, 6.6886, 7.1533, 7.7565, 9.4710	0

TABLE S21. Check of Theorem 8.1(i). The box multiplies friction by $[0.85, 1.15]$, C_{rr} by $[0.8, 1.3]$, $C_D A$ by $[0.9, 1.1]$, and P by $[0.9, 1.05]$. The columns give the envelope (m/s²) at the worst vertex, the smallest and largest envelope over n random points of the box, and the envelope at the best vertex.

Vehicle	Law	mph	n	Worst	Min	Max	Best
Leaf	ellipse	10	4096	3.2655	3.2680	3.8673	3.8723
Leaf	ellipse	60	4096	1.1250	1.1328	1.4542	1.4629
Leaf	PAC2002	10	512	3.2655	3.2710	3.8683	3.8723
Leaf	PAC2002	60	512	1.1250	1.1307	1.4433	1.4629
Civic	ellipse	10	4096	3.0622	3.0649	4.0383	4.0426
Civic	ellipse	60	4096	1.3910	1.3963	1.7685	1.7698
Civic	PAC2002	10	512	3.0256	3.0334	4.0091	4.0172
Civic	PAC2002	60	512	1.3910	1.4135	1.7543	1.7698
ZR1	ellipse	10	4096	6.1559	6.1645	9.1070	9.1154
ZR1	ellipse	60	4096	5.8688	5.9020	8.8581	8.8824
ZR1	PAC2002	10	512	5.9009	5.9305	8.5626	8.5651
ZR1	PAC2002	60	512	5.6255	5.6602	8.2991	8.3474

TABLE S22. Check of Theorem 8.1(iii) at 10 mph: the first-order standard deviation of $a_{\max}$, the Monte Carlo standard deviation over the 114 friction values (both m/s²), and the local amplification $m_{\text{eff}}/(m_{\text{eff}} - T')$ where traction binds.

Vehicle	Law	First order	Monte Carlo	Amplification
Leaf	ellipse	0.0000	0.0887	n/a
Leaf	PAC2002	0.0000	0.0922	n/a
Civic	ellipse	0.3813	0.3361	0.875
Civic	PAC2002	0.3855	0.3426	0.893
ZR1	ellipse	1.1573	1.0470	1.254
ZR1	PAC2002	1.0412	1.0019	1.187

S1.6. Minimum-time transit. Tables S23 and S24 give the values behind the check of Theorem 9.1.

TABLE S23. Check of Theorem 9.1 on the 6km rural route: the minimum trip time T^* (s) at selected multipliers on the nominal friction, from a sweep of 11 multipliers between 0.5 and 1.5 that is non-increasing in every case. The share of the route driven at the speed limit, at nominal friction with the ellipse law, is Leaf 95.0%, Civic 95.8%, ZR1 98.2%.

Vehicle	Law	$\times 0.5$	$\times 0.7$	$\times 1.0$	$\times 1.3$	$\times 1.5$
Leaf	ellipse	281.23	274.92	272.02	271.64	271.47
Leaf	PAC2002	281.30	274.94	272.02	271.64	271.47
Civic	ellipse	283.22	275.78	271.79	270.95	270.77
Civic	PAC2002	283.39	275.85	271.82	270.95	270.77
ZR1	ellipse	273.33	270.14	268.58	267.87	267.76
ZR1	PAC2002	273.44	270.25	268.68	267.87	267.76

TABLE S24. Trip-time quantiles (s) on the rural route with the ellipse law, computed as $T^*(q_\mu^+(1-p))$ from three solves at $p = 0.05, 0.5$, and 0.95 , and the largest difference from the Monte Carlo quantiles of the 114 solves at the measured friction values.

Vehicle	$p = 0.05$	$p = 0.5$	$p = 0.95$	Largest difference
Leaf	271.687	272.008	273.225	0
Civic	271.001	271.766	273.903	0
ZR1	267.907	268.555	269.463	0

S1.7. Safe transit. Table S25 gives the friction identified by each D3 drive cycle, Table S26 the values behind the check of Theorem 10.3, and Table S27 those behind the check of Proposition 10.5.

TABLE S25. Lower bound on friction identified from each Argonne D3 drive cycle, taking the larger of the driven-axle traction utilisation and the all-wheel braking utilisation, and in brackets its fraction of the median friction 0.97 pinned by the FMVSS 135 stops. The Focus Electric has no highway run.

Vehicle	UDDS (urban)	Highway	US06 (aggressive)
2010 Toyota Prius	0.32 (0.33)	0.32 (0.33)	0.72 (0.75)
2012 Ford Focus	0.36 (0.38)	0.35 (0.36)	0.69 (0.71)
2012 Nissan Leaf	0.36 (0.38)	0.29 (0.30)	0.60 (0.62)
2013 Ford Focus Electric	0.63 (0.65)	n/a	0.62 (0.64)
Median fraction	0.38	0.33	0.68

TABLE S26. Check of Theorem 10.3 on the rural route with the friction ellipse. For each vehicle and safety level δ , the plan is the minimum-time profile at the friction multiplier $q(\delta)$ of the 114 measured values, with trip time T^* (s). The plan was executed against all 114 values. The table gives the number of values below $q(\delta)$, the number at which the plan was not executable, and the infeasible share, which is at most δ in every case.

Vehicle	δ	$q(\delta)$	T^*	Below	Infeasible	Share
Leaf	0.05	0.805	273.23	5	5	0.044
Leaf	0.25	0.940	272.12	28	28	0.246
Civic	0.05	0.805	273.90	5	5	0.044
Civic	0.25	0.940	272.09	28	28	0.246
ZR1	0.05	0.805	269.46	5	5	0.044
ZR1	0.25	0.940	268.81	28	28	0.246

TABLE S27. Check of Proposition 10.5 on the urban route. For each vehicle and planning fraction r , the price (s) predicted by the transition-excess law, summed over the stop-go cycles, against the price computed by the three-pass solver, with their relative difference and the share of the predicted price due to braking.

Vehicle	r	Cycles	Law	Solver	Difference	Braking
Leaf	0.38	13	45.79	45.77	+0.0%	35%
Leaf	0.68	13	11.11	11.11	+0.0%	42%
Civic	0.38	13	54.97	54.94	+0.1%	31%
Civic	0.68	13	15.70	15.69	+0.1%	32%
ZR1	0.38	13	36.21	36.15	+0.1%	30%
ZR1	0.68	13	10.32	10.29	+0.3%	30%

S1.8. **The value of probing.** Table S28 gives the probe sequence of Proposition 11.1, and Table S29 the net gain of every number of probes behind Table 3.

TABLE S28. The sequence of slip-risk-limited probes of Proposition 11.1, with prior standard deviation $s_0 = 0.118$ of $\ln \mu$. For each probe the table gives its utilisation as a fraction of the friction and the standard deviation of $\ln \mu$ before and after it.

Probe	Utilisation	Before	After
1	0.824	0.1179	0.0312
2	0.950	0.0312	0.0126
3	0.979	0.0126	0.0078
4	0.987	0.0078	0.0059
5	0.990	0.0059	0.0048
6	0.992	0.0048	0.0041

S1.9. **Changing friction.** Table S30 gives the inputs of Theorem 12.1 for every vehicle and route, and compares the numerical minimiser of $J(D)$ with the asymptotic law at three

TABLE S29. Net trip-time gain (s) of k probes relative to planning at the 5% quantile of the fleet prior, for $k = 1, \dots, 5$, together with the price of planning at that prior, which is the largest gain possible. The optimal number k^* maximises the net gain, with $k^* = 0$ when every net gain is negative.

Vehicle	Route	Prior price	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	k^*
Leaf	highway	0.52	0.25	0.08	-0.14	-0.38	-0.62	1
Leaf	rural	0.95	0.65	0.42	0.14	-0.15	-0.44	1
Leaf	mountain	3.79	2.62	2.80	2.56	2.24	1.88	2
Leaf	urban	3.56	2.66	2.47	2.03	1.52	1.01	1
Civic	highway	1.01	0.60	0.55	0.38	0.18	-0.04	1
Civic	rural	1.82	1.37	1.28	1.05	0.80	0.54	1
Civic	mountain	4.43	3.09	3.43	3.25	2.98	2.67	2
Civic	urban	7.10	5.03	5.62	5.41	5.04	4.61	2
ZR1	highway	0.80	0.23	-0.13	-0.61	-1.12	-1.64	1
ZR1	rural	0.78	0.14	-0.34	-0.93	-1.56	-2.19	1
ZR1	mountain	0.57	-0.15	-0.82	-1.61	-2.41	-3.23	0
ZR1	urban	4.66	2.72	2.33	1.38	0.31	-0.80	1

correlation lengths. Table S31 gives the scaling of the numerical optimum with the correlation length.

TABLE S30. Inputs and results of Theorem 12.1, with field standard deviation $\sigma = 0.118$, $z = 1.645$, and probe residual standard deviation $s_p = 0.0323$. The columns give the probe cost c (s), the slope κ and the no-probe rate h_0 of the price curve (s/km), the threshold $\ell_{\min}$ (km), and for each correlation length ℓ the numerical minimiser of $J(D)$ and the asymptotic D^* (km). The entry none means that no spacing beats not probing. A dagger marks the cases within a factor of two of $\ell_{\min}$, which the threshold test excludes.

Vehicle	Route	c	κ	h_0	$\ell_{\min}$	$\ell = 1 \text{ km}$		$\ell = 5 \text{ km}$		$\ell = 25 \text{ km}$	
						Num.	Law	Num.	Law	Num.	Law
Leaf	highway	0.17	0.09	0.026	16.5	none	7.9	none	13.4	25.8 [†]	23.0
Leaf	rural	0.21	0.28	0.158	0.9	none [†]	4.1	4.1	7.0	8.0	11.9
Leaf	mountain	0.27	2.16	0.473	2.7	none	1.2	2.8 [†]	2.1	4.5	3.6
Leaf	urban	0.38	2.04	0.713	1.0	none [†]	1.6	2.3	2.7	5.7	4.7
Civic	highway	0.16	0.25	0.051	16.4	none	3.6	none	6.2	15.6 [†]	10.6
Civic	rural	0.19	0.81	0.303	1.0	none [†]	1.9	2.7	3.2	4.2	5.5
Civic	mountain	0.24	2.64	0.553	2.3	none	1.0	2.3	1.7	3.7	2.9
Civic	urban	0.34	6.69	1.421	1.2	1.5 [†]	0.7	1.5	1.2	2.5	2.0
ZR1	highway	0.37	0.19	0.040	46.6	none	7.6	none	13.1	none [†]	22.4
ZR1	rural	0.45	0.62	0.130	17.5	none	4.0	none	6.8	17.4 [†]	11.6
ZR1	mountain	0.57	0.33	0.071	39.9	none	7.1	none	12.1	none [†]	20.7
ZR1	urban	0.80	4.40	0.932	4.3	none	1.6	4.4 [†]	2.7	5.7	4.6

TABLE S31. Scaling of the numerical optimum with the correlation length for the Civic on the urban route, with a probe cost of 0.1 s and a residual standard deviation of 10^{-4} , close to the conditions of Theorem 12.1. The last column is the slope of $\ln D^*$ against $\ln \ell$, which the law predicts to be $1/3$.

ℓ_0 (km)	ℓ (km)	$D^*(\ell_0)$ (km)	$D^*(\ell)$ (km)	Slope
2	16	0.40	0.76	0.304
2	128	0.40	1.52	0.320

S1.10. **Real routes.** Table S32 describes the two real routes of Section 13, Table S33 decomposes the net gain of one probe on them, and Table S34 gives the check of Theorem 9.1.

TABLE S32. The two real routes after conversion: length (km), traffic signals, stop signs, accumulated climb of the smoothed elevation profile (m), share of route vertices with a posted-limit tag, and the tightest radius of the smoothed path (m).

Route	Length	Signals	Stop signs	Climb	Tagged	Tightest radius
Suburban	21.0	27	1	116	89%	21
Canyon	26.9	6	2	1120	54%	24

TABLE S33. One probe on the real routes. The prior price is T^* at the 5% quantile 0.824 of the fleet prior less $T^*(\mu)$. The gain is the reduction in T^* when one probe at 0.824 of the friction raises the planning fraction to 0.950, and the cost is $c(u_p)$ at the median posted limit of the route. All times are in seconds. The last column is the share of the prior price that the probe recovers.

Route	Vehicle	Prior price	Gain	Cost	Net	Recovered
Suburban, green	Leaf	1.04	0.86	0.30	0.56	54%
Suburban, green	Civic	1.69	1.29	0.27	1.03	61%
Suburban, green	ZR1	0.97	0.74	0.63	0.11	11%
Suburban, red	Leaf	9.37	7.93	0.30	7.63	81%
Suburban, red	Civic	18.85	14.22	0.27	13.95	74%
Suburban, red	ZR1	12.57	9.46	0.63	8.83	70%
Canyon, green	Leaf	0.76	0.64	0.26	0.38	50%
Canyon, green	Civic	1.47	1.12	0.24	0.88	60%
Canyon, green	ZR1	0.98	0.73	0.56	0.17	18%
Canyon, red	Leaf	2.32	1.98	0.26	1.72	74%
Canyon, red	Civic	4.65	3.50	0.24	3.27	70%
Canyon, red	ZR1	3.02	2.27	0.56	1.71	57%

TABLE S34. Check of Theorem 9.1 on the real routes: the minimum trip time T^* (s) of the Civic at selected multipliers on the nominal friction, from a sweep of 11 multipliers between 0.4 and 1.4 that is non-increasing in every case.

Route	$\times 0.4$	$\times 0.6$	$\times 0.8$	$\times 1.0$	$\times 1.2$	$\times 1.4$
Suburban, green	1246.8	1236.2	1232.3	1230.3	1229.1	1228.7
Suburban, red	1457.5	1380.1	1342.8	1320.8	1306.5	1303.1
Canyon, green	1512.0	1498.6	1495.3	1493.6	1492.5	1492.2
Canyon, red	1549.3	1524.4	1515.2	1509.8	1506.2	1505.4

S1.11. **Regime-resolved global sensitivity.** Table S37 gives the design of the Sobol computation of Section 14, Table S35 the first-order indices, Table S36 the total indices with the friction ellipse, and Table S38 the share of samples in which traction binds.

TABLE S35. First-order Sobol indices S_i with the PAC2002 tyre law, in the layout of Table 5. Negative estimates are shown as zero. The largest bootstrap standard error of any first-order index is 0.090.

Input	Leaf			Civic			ZR1		
	10	60	Pool	10	60	Pool	10	60	Pool
Friction	0.21	0.00	0.00	0.23	0.00	0.00	0.35	0.09	0.13
Wetness	0.22	0.72	0.03	0.30	0.59	0.05	0.48	0.80	0.55
Wet-to-dry ratio	0.02	0.00	0.00	0.01	0.00	0.00	0.02	0.00	0.00
Temperature	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00
Altitude	0.00	0.01	0.00	0.22	0.09	0.02	0.00	0.03	0.02
C_{rr}	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C_{DA}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Headwind	0.02	0.05	0.00	0.00	0.05	0.00	0.00	0.00	0.00
Payload	0.16	0.02	0.01	0.02	0.03	0.00	0.00	0.01	0.00
Grade	0.31	0.22	0.02	0.13	0.14	0.02	0.01	0.00	0.02

TABLE S36. Total Sobol indices ST_i with the friction ellipse, in the layout of Table 5. No entry differs from the PAC2002 value by more than 0.06.

Input	Leaf			Civic			ZR1		
	10	60	Pool	10	60	Pool	10	60	Pool
Friction	0.31	0.00	0.01	0.27	0.00	0.02	0.43	0.09	0.16
Wetness	0.33	0.71	0.05	0.33	0.64	0.07	0.52	0.88	0.71
Wet-to-dry ratio	0.05	0.00	0.00	0.04	0.00	0.00	0.05	0.01	0.02
Temperature	0.00	0.00	0.00	0.03	0.01	0.00	0.01	0.01	0.01
Altitude	0.00	0.01	0.00	0.40	0.14	0.04	0.15	0.07	0.08
C_{rr}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C_DA	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Headwind	0.01	0.06	0.00	0.00	0.04	0.00	0.00	0.00	0.00
Payload	0.24	0.03	0.01	0.08	0.03	0.01	0.03	0.01	0.01
Grade	0.32	0.20	0.02	0.13	0.17	0.02	0.02	0.01	0.01
Speed	0.91			0.89			0.23		

TABLE S37. Design of the Sobol computation. The inputs are independent and uniform on the stated ranges. Each index uses $N = 8192$ rows per sample matrix, and the whole computation used 1,818,624 evaluations of the envelope. Standard errors are from 200 bootstrap resamples of the rows, and the largest for any total index is 0.019.

Input	Range
Friction multiplier	[0.81, 1.24]
Wetness w , dry for $w < 0.5$, film $(w - 0.5) \times 6$ mm otherwise	[0, 1]
Wet-to-dry peak friction ratio	[0.71, 0.87]
Ambient temperature ($^{\circ}\text{C}$)	[−10, 40]
Altitude (m)	[0, 3000]
C_{rr} multiplier	[0.8, 1.3]
C_DA multiplier	[0.9, 1.1]
Headwind (m/s)	[−5, 15]
Payload (kg)	[0, 300]
Uphill grade (%)	[0, 6]

TABLE S38. Share of the first 200 samples of each Sobol design in which traction binds.

Vehicle	Tyre law	10 mph	60 mph
Leaf	ellipse	0.31	0.09
Leaf	PAC2002	0.28	0.10
Civic	ellipse	0.54	0.08
Civic	PAC2002	0.52	0.06
ZR1	ellipse	0.66	0.54
ZR1	PAC2002	0.76	0.61

S1.12. **Transient deficit.** Tables S41 and S42 give the penalties of the three vehicles in Section 15, and Tables S39 and S40 the checks of Theorems 15.1 and 15.2 and of Proposition 15.3.

TABLE S39. Checks of Theorems 15.1 and 15.2 on random piecewise-linear envelopes. The non-increasing envelopes have seven levels on random knots, lags drawn uniformly from $[0.02, 0.4]$ s, and target speeds from $[5, 40]$ m/s. The general envelopes have eight random levels and one lag drawn from $[0.05, 0.4]$ s. The last two columns give the median and the largest ratio of the penalty to the stated bound. The last row compares the general envelopes with the bound τ that holds for non-increasing envelopes.

Envelope	Lags	Cases	Bound	Median ratio	Largest ratio
Non-increasing	1	18	$\sum_j \tau_j$	0.881	0.987
Non-increasing	2	17	$\sum_j \tau_j$	0.832	0.986
Non-increasing	3	25	$\sum_j \tau_j$	0.869	0.971
General	1	40	$\tau(1 + \text{Var}^+)$	0.644	0.903
General	1	40	τ	2.314	4.033

TABLE S40. Penalty of a single lag $\tau = 0.2$ s on a doubling staircase envelope, in units of τ , with each level held for about 5τ . The prediction is the limit of Proposition 15.3, and the bound is that of Theorem 15.2.

Levels	Penalty	Prediction	Bound
1	1.00	1.00	1.00
2	1.50	1.50	1.69
4	2.49	2.50	3.08
6	3.49	3.50	4.47
8	4.49	4.50	5.85

TABLE S41. The 0–60 mph penalty of the three vehicles under constant lags (s), with the time T_{phys} along the envelope and the penalty as a share of $\sum_j \tau_j$. The positive variation of $\ln a_{\text{max}}$ from rest to 60 mph, computed on 4001 speeds, is 0 for every vehicle.

Vehicle	Lags (s)	T_{phys} (s)	Penalty (s)	Share
Leaf	0.10	10.572	0.100	99.8%
Leaf	0.15, 0.05	10.572	0.199	99.6%
Leaf	0.30, 0.10, 0.05	10.572	0.448	99.5%
Civic	0.10	8.964	0.100	99.8%
Civic	0.15, 0.05	8.964	0.199	99.7%
Civic	0.30, 0.10, 0.05	8.964	0.448	99.5%
ZR1	0.10	3.578	0.100	99.6%
ZR1	0.15, 0.05	3.578	0.199	99.4%
ZR1	0.30, 0.10, 0.05	3.578	0.446	99.1%

TABLE S42. The 0–60 mph penalty (s) with tyre relaxation length 0.5 m and a powertrain lag of 0.10 s, against the floor speed v_0 of the relaxation model.

Vehicle	$v_0 = 2 \text{ m/s}$	$v_0 = 1 \text{ m/s}$	$v_0 = 0.5 \text{ m/s}$	$v_0 = 0.25 \text{ m/s}$
Leaf	0.346	0.540	0.773	1.021
Civic	0.347	0.542	0.778	1.029
ZR1	0.335	0.486	0.654	0.829

S2. PROOFS FOR THE TRANSIENT DEFICIT

In this section we prove Theorems 15.1 and 15.2 and Proposition 15.3 of the main text. We first write the penalty as an integral, and then prove two elementary facts on which all three proofs rest.

Along the lagged run we write $v(t)$ for the speed and $A(t) = a_{\max}(v(t))$ for the envelope at that speed. The states $y_1, \dots, y_k$ of the cascade start from rest and satisfy

$$y'_j = \frac{y_{j-1} - y_j}{\tau_j}, \quad y_j(0) = 0, \quad j = 1, \dots, k,$$

with input $y_0 = A$. The delivered acceleration is $d = \min(y_k, A)$, so that $v' = d$. Each stage is a weighted average of its input,

$$y_j(t) = \frac{1}{\tau_j} \int_0^t e^{-(t-s)/\tau_j} y_{j-1}(s) ds,$$

so every y_j is non-negative because A is positive. Hence $d \geq 0$, and the speed is non-decreasing.

Lemma S2.1 (The penalty as an integral). *Let the lagged run reach the target speed V at the time T_{real} . Then*

$$(S1) \quad T_{\text{real}} - T_{\text{phys}} = \int_0^{T_{\text{real}}} \left(1 - \frac{d(t)}{A(t)}\right) dt.$$

Proof. The time along the envelope is $T_{\text{phys}} = \int_0^V dw/a_{\max}(w)$. The speed v is non-decreasing, with $v(0) = 0$, $v(T_{\text{real}}) = V$, and $v' = d$, so the substitution $w = v(t)$ gives

$$T_{\text{phys}} = \int_0^{T_{\text{real}}} \frac{d(t)}{a_{\max}(v(t))} dt = \int_0^{T_{\text{real}}} \frac{d(t)}{A(t)} dt.$$

Subtracting this from $T_{\text{real}} = \int_0^{T_{\text{real}}} dt$ gives (S1). Since $d \leq A$, the integrand is non-negative, so the penalty is non-negative. $\square$

The integrand of (S1) is the shortfall of the delivered acceleration relative to the envelope. The proofs below bound it with two elementary facts. The first is a comparison principle for a single lag, and the second describes the response of the cascade to a unit step.

Lemma S2.2 (Comparison for one lag). *Let $\tau > 0$, and let e be absolutely continuous on $[t_0, t_1]$ with $e' \geq -e/\tau$ almost everywhere. Then $e(t) \geq e(t_0) e^{-(t-t_0)/\tau}$ for every $t \in [t_0, t_1]$. In particular, $e(t_0) \geq 0$ implies $e \geq 0$ on $[t_0, t_1]$.*

Proof. The function $e^{t/\tau} e(t)$ has derivative $e^{t/\tau} (e' + e/\tau) \geq 0$, so it is non-decreasing. Hence $e^{t/\tau} e(t) \geq e^{t_0/\tau} e(t_0)$, which is the claim. $\square$

Lemma S2.3 (Step response of the cascade). *Let $\zeta_0 \equiv 1$, and let $\zeta_1, \dots, \zeta_k$ satisfy $\zeta'_j = (\zeta_{j-1} - \zeta_j)/\tau_j$ with $\zeta_j(0) = 0$. Then each ζ_j is non-decreasing, with $0 \leq \zeta_j \leq 1$ and $\zeta_j(t) \rightarrow 1$ as $t \rightarrow \infty$, and*

$$\int_0^\infty (1 - \zeta_k(t)) dt = \sum_{j=1}^k \tau_j.$$

Proof. We argue by induction on j , the case $j = 0$ being trivial. Suppose that ζ_{j-1} is non-decreasing, takes values in $[0, 1]$, and tends to 1. As above,

$$\zeta_j(t) = \frac{1}{\tau_j} \int_0^t e^{-(t-s)/\tau_j} \zeta_{j-1}(s) ds.$$

This is non-negative, and since $\zeta_{j-1}(s) \leq \zeta_{j-1}(t)$ for $s \leq t$, it is at most $(1 - e^{-t/\tau_j})\zeta_{j-1}(t) \leq \zeta_{j-1}(t) \leq 1$. Therefore $\zeta'_j = (\zeta_{j-1} - \zeta_j)/\tau_j \geq 0$, so ζ_j is non-decreasing. It tends to 1 because it is a stable first-order response to an input that tends to 1.

For the integral, note that $\zeta_{j-1} - \zeta_j = \tau_j \zeta'_j \geq 0$, so

$$\int_0^\infty (\zeta_{j-1} - \zeta_j) dt = \tau_j \lim_{t \rightarrow \infty} (\zeta_j(t) - \zeta_j(0)) = \tau_j.$$

The differences telescope, $1 - \zeta_k = \sum_{j=1}^k (\zeta_{j-1} - \zeta_j)$, and summing over j gives the result. $\square$

Proof of Theorem 15.1. We compare the cascade along the run with the step response of Lemma S2.3. Let $z_j = y_j/A$ for $j = 0, \dots, k$, so that $z_0 \equiv 1$. The function A is absolutely continuous, because $a_{\max}$ is locally Lipschitz and v is absolutely continuous. Differentiating z_j gives

$$\begin{aligned} z'_j &= \frac{y'_j}{A} - \frac{y_j A'}{A^2} \\ &= \frac{z_{j-1} - z_j}{\tau_j} - z_j \frac{A'}{A}. \end{aligned}$$

The speed is non-decreasing and $a_{\max}$ is non-increasing, so A is non-increasing and $A' \leq 0$ almost everywhere. Since $z_j \geq 0$, the last term is non-negative, and

$$(S2) \quad z'_j \geq \frac{z_{j-1} - z_j}{\tau_j}.$$

We now show that $z_j \geq \zeta_j$ by induction on j . Both sides equal 1 for $j = 0$. Suppose that $z_{j-1} \geq \zeta_{j-1}$. By (S2), $z'_j \geq (\zeta_{j-1} - z_j)/\tau_j$, while $\zeta'_j = (\zeta_{j-1} - \zeta_j)/\tau_j$. The difference $e = z_j - \zeta_j$ therefore satisfies $e' \geq -e/\tau_j$ with $e(0) = 0$, and Lemma S2.2 gives $e \geq 0$.

In particular $z_k \geq \zeta_k$, and $\zeta_k \leq 1$ by Lemma S2.3. The integrand of (S1) therefore satisfies

$$1 - \frac{d}{A} = 1 - \min(z_k, 1) \leq 1 - \zeta_k.$$

By Lemmas S2.1 and S2.3,

$$T_{\text{real}} - T_{\text{phys}} \leq \int_0^{T_{\text{real}}} (1 - \zeta_k) dt \leq \int_0^\infty (1 - \zeta_k) dt = \sum_{j=1}^k \tau_j.$$

For a constant envelope $A' = 0$, so (S2) holds with equality and $z_k = \zeta_k \leq 1$. The penalty is then exactly $\int_0^{T_{\text{real}}} (1 - \zeta_k) dt$, which tends to $\sum_j \tau_j$ as T_{real} grows with the target speed. $\square$

Proof of Theorem 15.2. With one lag we write $y = y_1$, and we define the *relative debt* $r = 1 - y/A$. It starts at $r(0) = 1$, and $r \leq 1$ because $y \geq 0$. Since $d/A = \min(y/A, 1)$, the integrand of (S1) is

$$1 - \frac{d}{A} = \max(r, 0).$$

Step 1: a differential inequality for the debt. Differentiating r gives

$$\begin{aligned} r' &= -\frac{y'}{A} + \frac{y A'}{A^2} \\ &= -\frac{A-y}{\tau A} + (1-r)\frac{A'}{A} \\ &= -\frac{r}{\tau} + (1-r)\frac{A'}{A}. \end{aligned}$$

Let $\gamma = \max(A'/A, 0)$. Wherever $0 \leq r \leq 1$, the factor $1-r$ lies in $[0, 1]$, so $(1-r)A'/A \leq \gamma$, and

$$(S3) \quad r' \leq \gamma - \frac{r}{\tau}.$$

Step 2: comparison with a majorant. Let ρ solve $\rho' = \gamma - \rho/\tau$ with $\rho(0) = 1$, that is,

$$\rho(t) = e^{-t/\tau} + \int_0^t e^{-(t-s)/\tau} \gamma(s) ds.$$

Both terms are non-negative, so $\rho \geq 0$. We claim that $\max(r, 0) \leq \rho$. Since $\rho \geq 0$, it suffices to show that $r \leq \rho$ wherever $r > 0$. Suppose instead that $r(t_1) > \rho(t_1)$ for some t_1 , and let t_0 be the last time before t_1 at which $r \leq \rho$. Such a time exists because $r(0) = \rho(0)$, and $r(t_0) = \rho(t_0)$ by continuity. On $(t_0, t_1]$ we have $r > \rho \geq 0$, so (S3) holds there, and $e = \rho - r$ satisfies $e' \geq -e/\tau$ with $e(t_0) = 0$. Lemma S2.2 gives $e(t_1) \geq 0$, which contradicts $r(t_1) > \rho(t_1)$.

Step 3: integration. Extend γ by zero after T_{real} . By Lemma S2.1 and Step 2,

$$T_{\text{real}} - T_{\text{phys}} \leq \int_0^{T_{\text{real}}} \rho dt \leq \int_0^\infty \rho dt = \tau + \tau \int_0^{T_{\text{real}}} \gamma dt,$$

where the last step exchanges the order of integration and uses $\int_0^\infty e^{-t/\tau} dt = \tau$.

Step 4: the variation. By the chain rule, $(\ln A)' = (\ln a_{\text{max}})'(v) v'$, and $v' \geq 0$, so $\gamma = \max((\ln a_{\text{max}})'(v), 0) v'$. The substitution $w = v(t)$ then gives $\int_0^{T_{\text{real}}} \gamma dt = \text{Var}^+$, which completes the proof. $\square$

Proof of Proposition 15.3. The relative debt $r = 1 - y/A$ of the proof of Theorem 15.2 is again the integrand of (S1) wherever $r \geq 0$. On each level A is constant, so $r' = -r/\tau$, and r decays as $e^{-t/\tau}$. A level on which r starts at $r_i \geq 0$ and which lasts the time h_i therefore contributes $\tau r_i(1 - e^{-h_i/\tau})$ to the penalty, and this tends to τr_i as h_i grows.

The first level starts at $r_0 = 1$ and contributes τ in the limit. At the step from a_{i-1} to a_i , the state y is continuous while A jumps, so the debt jumps to

$$r_i = 1 - \frac{y}{a_i} = 1 - (1 - r(t_i^-)) \frac{a_{i-1}}{a_i},$$

where $r(t_i^-)$ is the debt just before the step. Since $a_{i-1} < a_i$, this gives $r_i \geq 1 - a_{i-1}/a_i > 0$ whenever $r(t_i^-) \geq 0$, so by induction the debt is positive throughout the run. As the time on level $i-1$ grows, $r(t_i^-) \rightarrow 0$, and so $r_i \rightarrow 1 - a_{i-1}/a_i$. Summing the contributions of all levels gives the limit. For $a_i = a_0 q^i$, every step contributes $1 - 1/q$. $\square$

Proof of Remark 15.4. With $\tau = \sigma/v$ the lag runs in distance rather than in time. Writing x for the distance travelled and $a > 0$ for the envelope near rest, the lag state satisfies $dy/dx = (a - y)/\sigma$, so $y \approx ax/\sigma$ for small x . Since $v dv/dx = y$, this gives $v \approx x\sqrt{a/\sigma}$ near

rest. The time $\int dx/v$ to cover any distance from rest therefore diverges logarithmically at $x = 0$. The floor speed v_0 removes this divergence, and the penalty grows without bound as $v_0 \rightarrow 0$. $\square$

REFERENCES

- [1] Corvette Action Center, *2013 Corvette ZR1 specifications*, accessed 2026, https://www.corvetteactioncenter.com/specs/c6/2013/2013_zr1_specs.html.
- [2] EV Specifications, *2013 Nissan Leaf specifications*, accessed 2026, <https://www.evspecifications.com/en/model/7b505>.
- [3] MotorWeek, *Road test: 2013 Honda Civic*, accessed 2026, https://motorweek.org/road_tests/2013_honda_civic/.